

EM Propagation in General Materials:- (σ, ϵ, μ)

Using Maxwell's Equation

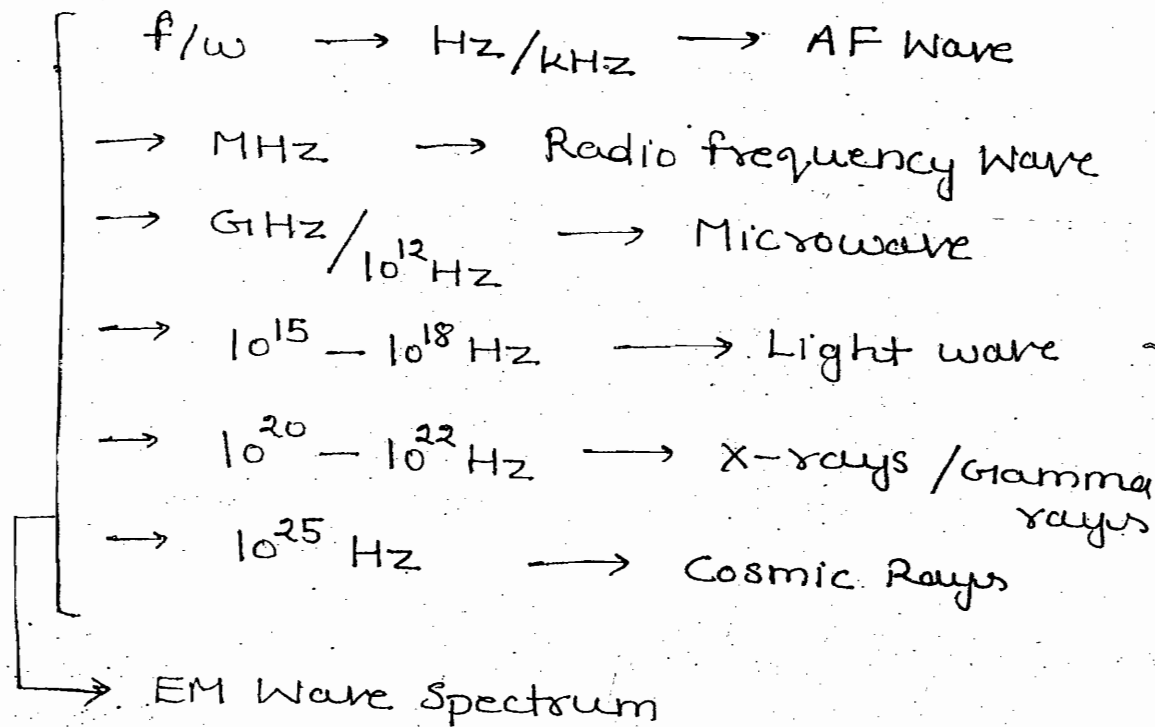
$$\nabla \times H = \sigma E + \epsilon \frac{\partial E}{\partial t}$$

$$\nabla \times E = -\mu \frac{\partial H}{\partial t}$$

→ Every EM wave needs a time harmonic source at one end

$$\left. \begin{aligned} E_s &= E_0 e^{j\omega t} \\ H_s &= H_0 e^{j\omega t} \end{aligned} \right\} \omega = \text{frequency in rad/s}$$

Range of ω :-



$$\nabla \times H = \sigma E + \epsilon \frac{\partial E}{\partial t} = \sigma E_0 e^{j\omega t} + \epsilon j\omega E_0 e^{j\omega t}$$

$$\nabla \times E = -\mu \frac{\partial H}{\partial t} = (\sigma + j\omega\epsilon) E$$

$$\Rightarrow \nabla \times H = (\sigma + j\omega\epsilon) E \quad \text{--- (1)}$$

$$\nabla \times E = -j\omega\mu H \quad \text{--- (2)}$$

$$\begin{aligned}\nabla \times E &= \mu \frac{\partial H}{\partial t} \\ &= -\mu j\omega H_0 e^{j\omega t} = -j\omega\mu H\end{aligned}$$

$$\nabla \times E = -j\omega\mu H \quad \text{--- (1)}$$

Eq- (1) & (11) are called as Maxwell Equations for time Harmonic source

Take H from (11) & put in (1)

$$H = \frac{\nabla \times E}{-j\omega\mu}$$

$$\Rightarrow \nabla \times \left(\frac{\nabla \times E}{-j\omega\mu} \right) = (\sigma + j\omega\epsilon) E$$

$$\Rightarrow \nabla \cdot \left(\underbrace{\nabla \cdot E}_{=0} \right) - \nabla^2 E = -j\omega\mu (\sigma + j\omega\epsilon) E$$

For a charge free region

$$\nabla^2 E = j\omega\mu (\sigma + j\omega\epsilon) E \quad \text{--- (11)}$$

Similarly

$$\underbrace{\nabla^2 H}_{\text{Space}} = j\omega\mu (\sigma + j\omega\epsilon) \underbrace{H}_{\text{time}} \quad \text{--- (1V)}$$

These are called as Helmholtz's Propagation Equations.

Note:-

→ If the source is the time harmonic then the effect is space harmonic propagation in the medium.

Lecture - 9

$$\nabla^2 E = j\omega\mu (\sigma + j\omega\epsilon) E \quad \text{--- (III)}$$

$$\nabla^2 H = j\omega\mu (\sigma + j\omega\epsilon) H \quad \text{--- (IV)}$$

Consider ∇ in one dimension dimension of z only
 E is in x direction, H is in y -direction

$$\text{Consider } \sqrt{j\omega\mu (\sigma + j\omega\epsilon)} = \gamma$$

$$\frac{\partial^2 E_x}{\partial z^2} = \gamma^2 E_x \quad \text{--- (III)}$$

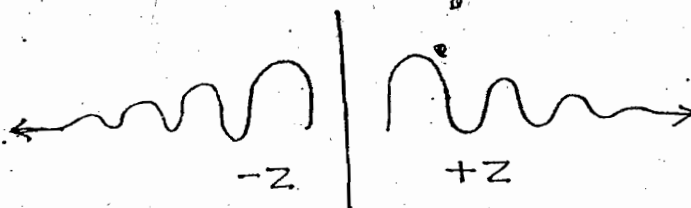
$$\frac{\partial^2 H_y}{\partial z^2} = \gamma^2 H_y \quad \text{--- (IV)}$$

The $E(z)$ and $H(z)$ solution is.

$$E(z)_x = (c_1 e^{-\gamma z} + c_2 e^{\gamma z}) a_x \quad \text{--- (V)}$$

$$H(z)_y = (c_3 \underset{\substack{\downarrow \\ +z}}{e^{-\gamma z}} + c_4 \underset{\substack{\downarrow \\ -z}}{e^{\gamma z}}) a_y \quad \text{--- (VI)}$$

The solution physically represent two waves travelling on either sides from a planar source and both decaying. This is called as uniform plane wave.



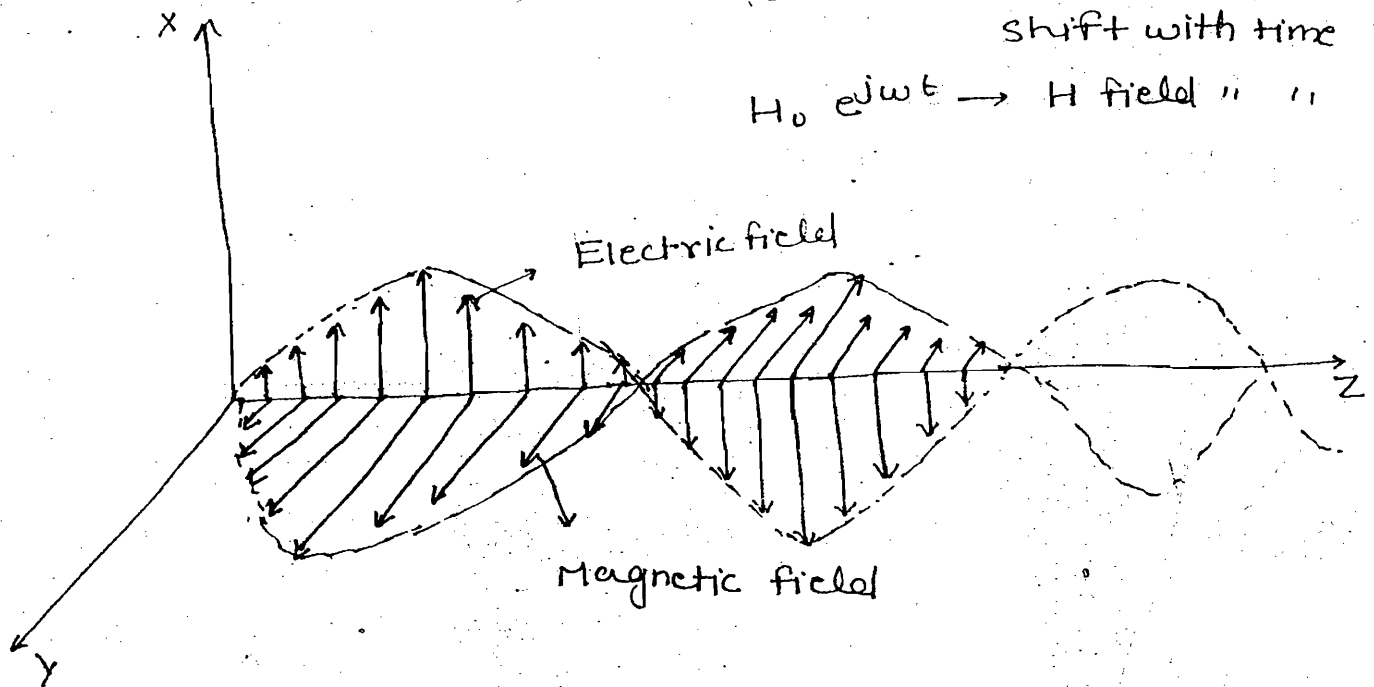
Considering only the $+z$ solution and c_1 & c_3 being the initial source values. The final E/H wave solution is

$$E(z, t)_x = (E_0 e^{j\omega t} \cdot e^{-\gamma z}) a_x$$

$$H(z, t)_y = (H_0 e^{j\omega t} \cdot e^{-\gamma z}) a_y$$

The solution is a product solution of time and space harmonics.

$E_0 e^{j\omega t} \rightarrow$ electric field shift with time
 $H_0 e^{j\omega t} \rightarrow$ H field " "



- Electric and H field perpendicular and shifts simultaneously with time
- Cross product result in transverse and dot product result in longitudinal.

Propagation constant (γ) :-

γ is the constant of the exponential with z that decides the course of propagation i.e. γ is called as propagation constant

$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)} = \alpha + j\beta$$

$$E(z, t) = (E_0 e^{-\alpha z} \cdot e^{j\omega t} e^{-j\beta z}) a_x$$

$$E(z, t)_x = (E_0 e^{-\alpha z}) e^{j(\omega t - \beta z)} a_x$$

$$H(z, t)_y = (H_0 e^{-\alpha z}) e^{j(\omega t - \beta z)} a_y$$

Field direction

Amplitude Phase

Note-1:-

Every EM wave is harmonic whose amplitude exponentially decays at α rate

α = attenuation constant

Note-2:-

Every EM wave is a harmonic whose phase independently but linearly changes with time and space.

Note-3:-

The harmonic E & H have direction and obeying the basic transverse nature with propagation

$$\begin{array}{ccc} E & \times & H \\ (\text{Dir}) & & (\text{Dir}) \end{array} = \begin{array}{c} \text{Propagation constant} \\ (\text{Dir}) \end{array}$$

Intrinsic Wave Impedance (η) :-

$$E \Rightarrow H$$

$$\eta = \frac{E}{H} = \frac{\text{Volts/m}}{\text{Amp/m}} = \text{ohm}$$

When E is transformed to H & vice-versa the rate of transformation is called as intrinsic impedance of the medium.

$$\nabla \times E = -j\omega\mu H$$

$$\frac{\partial}{\partial z} a_z \times E_{ax} = -j\omega\mu H a_y$$

$$\frac{\partial E}{\partial z} = -j\omega\mu H$$

$$\text{As } E = E_0 e^{-\gamma z}$$

$$\Rightarrow -\gamma E = -j\omega\mu H$$

$$\Rightarrow \frac{E}{H} = \frac{j\omega\mu}{\sqrt{j\omega\mu(\sigma + j\omega\epsilon)}}$$

$$\Rightarrow \boxed{\frac{E}{H} = \sqrt{\frac{j\omega\mu}{1 + j\omega\epsilon}}}$$

$$\eta = \sqrt{\frac{j\omega\mu}{1 + j\omega\epsilon}} = \text{complex} = R + jX$$

The transformation has a resistance which has a loss (α) in transformation and a reactance which leads to phase shift (β)

$$Y = \alpha + j\beta$$

$$\eta = \underset{\uparrow}{R} + j\underset{\uparrow}{X}$$

Workbook-2

$$1) \quad V = -\frac{d\psi_m}{dt} = -\frac{d}{dt} \left(\frac{1}{3} t^3 \right) = -\frac{1}{3} \cdot 3t^2 \Big|_{t=3s} = 9V$$

$$\Rightarrow \boxed{\lambda = -1}$$

$$\begin{aligned} (2) \quad V &= -\frac{d\psi_m}{dt} = -\frac{d}{dt} (BA) = -\pi r^2 \frac{dB}{dt} \\ &= -\pi \times (0.1)^2 \times 10 \left[-\sin(120\pi t) \cdot 120\pi \right] \\ &= 118 \sin(120\pi t) \quad \text{Ans} \end{aligned}$$

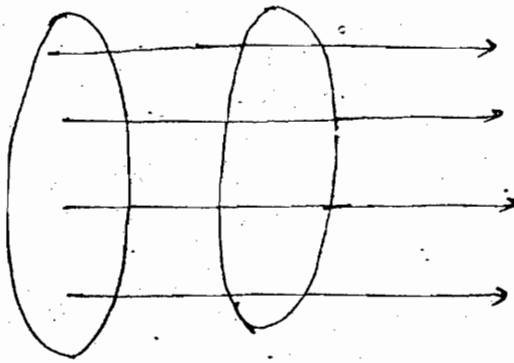
3) $B \rightarrow$ Linearly decrease with time $B(t) = -kt$
 $\rightarrow$ Uniform decrease with space

$$V = -\frac{d\psi_m}{dt} = -\frac{d}{dt} (BA) = -A(-k) = \text{constant}$$

= DC voltage

$\rightarrow$ Upto open circuit E flows and emf is induced

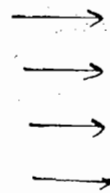
$\rightarrow$ Induced emf converts into current in coil 1



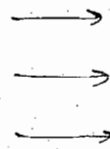
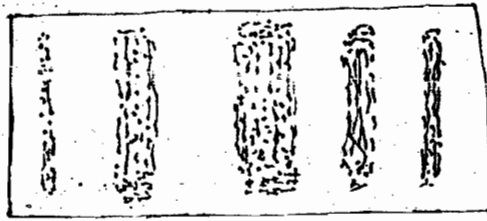
Ans - D

$$\begin{array}{ll} V=0 & V \neq 0 \\ I \neq 0 & I=0 \end{array}$$

Note:-



DC current



AC current

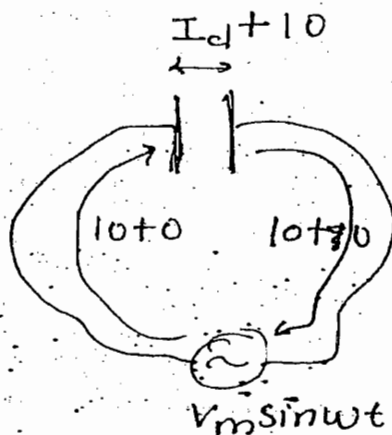
1.4.

$$I_c = I_d$$

$$\oint \mathbf{H} \cdot d\mathbf{l} = I_c + I_d \neq 2I_d \text{ or } 2I_c$$

$$J_d = \epsilon \frac{\partial E}{\partial t}$$

$$I_d = \epsilon A \cdot j\omega E = j\omega \frac{\epsilon A}{d} V$$



Ans - (a)

$$\vec{B(t)} \times$$

Uniform and Harmonic
in time

Transformer Action

involved

$$V = - \frac{d\psi_m}{dt} = - \frac{d(BA)}{dt} = - B \underbrace{\frac{dA}{dt}}_{\text{Rotation}} - A \frac{dB}{dt}$$

$$B(t) = \cos(\omega t) \rightarrow \text{AC voltage}$$

Ans - c)

Radiation $\rightarrow$ EM Waves $\rightarrow$ generate due to cross product

$$Y = \alpha + j\beta$$

$$\text{Amplitude} = |E|$$

$$|E| = E_0 e^{-\alpha z} = \frac{1}{e} E_0 \quad \text{when } z = 20\text{m}$$

$$\Rightarrow \alpha z = 1 \Rightarrow \alpha = \frac{1}{20}$$

$$\theta = \alpha z \Rightarrow \theta = \beta z$$

$$\Rightarrow \beta = \frac{\pi/6}{20\text{m}} = \frac{\pi}{120}$$

$$Y = \frac{1}{20} + j \frac{\pi}{120}$$

$|E| \rightarrow 20\%$ of initial value

$$|E| \rightarrow 20 \xrightarrow{E_0} 100 \rightarrow z = 5\text{m}$$

$$|E| \rightarrow 40 \xrightarrow{100} z = ?$$

It is not a linear function

$$|E| = E_0 e^{-\alpha z}$$

$$\Rightarrow 20 = 100 e^{-\alpha 5} \Rightarrow e^{5\alpha} = 5$$

$$\Rightarrow 5\alpha = \ln 5$$

$$\Rightarrow \alpha = \frac{\ln 5}{5}$$

$$40 = 100 e^{-\alpha z}$$

$$\Rightarrow e^{\alpha z} = \frac{100}{40} = 2.5$$

$$\Rightarrow \alpha z = \ln 2.5$$

$$\Rightarrow z = \frac{5 \ln 2.5}{\ln 5} = 2.8, \text{Ans.}$$

Note:-

Change is depending on function value

Case-(1):-

$$|E| = E_0 e^{-\alpha z}$$

EM Wave
Propagation
in free
space

$\frac{3}{\alpha}$

$$\left[\begin{array}{l} E_0 = 100 \\ \downarrow \\ z = \frac{1}{\alpha} \rightarrow \frac{100}{e} = 37 \\ \downarrow \\ z = \frac{1}{\alpha} \rightarrow \frac{100}{e^2} = 13 \\ \downarrow \\ z = \frac{1}{\alpha} \rightarrow \frac{100}{e^3} = 4 \end{array} \right]$$

Case - (1) :-

EM Wave Propagation in free space $\left[\sigma = 0, \epsilon = \epsilon_0 \right]$
 $\mu = \mu_0$

$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)} = j\omega\sqrt{\mu_0\epsilon_0} = \text{purely imaginary}$$

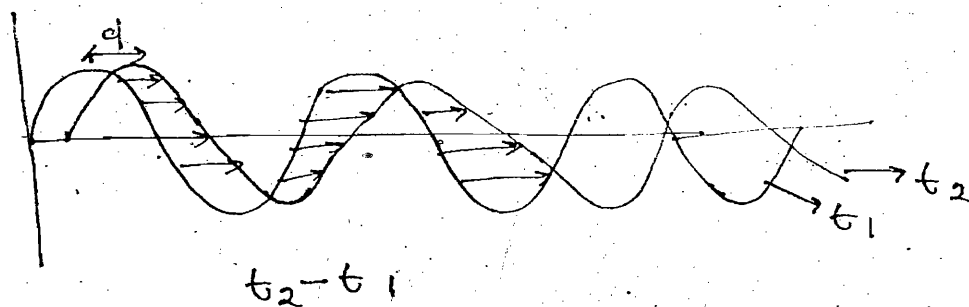
$\Rightarrow \alpha = 0$ No attenuation of EM wave in free space

$\beta = \omega\sqrt{\mu_0\epsilon_0}$ = only phase shift and propagation constant exists

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}} = \sqrt{\frac{\mu_0}{\epsilon_0}} = \text{real}$$

$$= \sqrt{4\pi \times 10^{-7} \times 36\pi \times 10^9} = 120\pi = 377$$

Phase Velocity (V_p) :-



The distance travelled by any in-phase point per unit time is called as phase velocity.

$$V_p = \frac{d}{t}$$

$$= \frac{\lambda}{T} = \lambda f = \frac{2\pi\lambda}{2\pi T}$$

$$\Rightarrow V_p = \frac{\omega}{\beta} = \frac{\omega}{\omega\sqrt{\mu_0\epsilon_0}} = \frac{1}{\sqrt{\mu_0\epsilon_0}}$$

$$\Rightarrow V_p = 3 \times 10^8 \text{ m/s}$$

Case - (ii) :-

EM Wave propagation in Ideal dielectrics

→ Ideal dielectrics = Perfect / lossless dielectrics

$$\rightarrow \sigma = 0, \epsilon = \epsilon_0 \epsilon_r, \mu = \mu_0$$

$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)} = j\omega\sqrt{\mu_0\epsilon_0\epsilon_r}$$

= purely imaginary

$\Rightarrow \alpha = 0$ i.e. No attenuation of EM wave in ideal dielectrics

$$\beta = \omega\sqrt{\mu_0\epsilon_0\epsilon_r} = \text{only phase shift and property exist}$$

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}} = \sqrt{\frac{\mu_0}{\epsilon_0\epsilon_r}} = \text{real}$$

$$= \sqrt{\frac{4\pi \times 10^{-7} \times 36\pi \times 10^9}{\epsilon_r}} = \frac{120\pi}{\sqrt{\epsilon_r}} = \frac{377}{\sqrt{\epsilon_r}}$$

Phase Velocity (V_p) :-

$$V_p = \frac{\lambda}{T} = \lambda f = \frac{2\pi\lambda}{2\pi T}$$

$$\Rightarrow V_p = \frac{\omega}{\beta} = \frac{\omega}{\omega\sqrt{\mu_0\epsilon_0}} = \frac{1}{\sqrt{\mu_0\epsilon_0\epsilon_r}}$$

$$\Rightarrow \boxed{V_p = \frac{3 \times 10^8}{\sqrt{\epsilon_r}} \text{ m/s}}$$

$$E(x, t) = 25 \sin(\omega t + 4x) a_y$$

By comparison, propagation direction $\rightarrow -a_x$

$$\beta = 4 = \frac{2\pi}{\lambda} \Rightarrow \lambda = \frac{\pi}{2}$$

$$f = \frac{3 \times 10^8}{\frac{\pi}{2}} = \frac{600}{\pi} \text{ MHz}$$

Method-1:-

$$\nabla \times E = -j\omega\mu H = -\mu \frac{\partial H}{\partial t}$$

$$H = -\frac{1}{\mu} \int (\nabla \times E) dt$$

$$\nabla \times E = \begin{vmatrix} a_x & a_y & a_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & E_y & 0 \end{vmatrix}$$

Method 2:-

$$E(x, t) = 25 \sin(\omega t + 4x) a_y$$

Step-1:- $\frac{|E|}{|H|} = |n|$

Step-2:- when n is real E & H have the same harmonic and phase

Step-3:- $\frac{E}{\text{div}} \times \frac{H}{\text{div}} = \frac{P_{\text{avg}}}{\text{div}}$ (Basic Transverse Nature)

$$\begin{aligned} H(x, t) &= \left(\frac{25}{120\pi} \right) \sin(\omega t + 4x) (-a_x \times a_y) \\ &= \left(\frac{-25}{120\pi} \right) \sin(\omega t + 4x) a_z \end{aligned}$$

$$E(y, t) = 25 \sin(10^8 t - y) a_z$$

const
free space lossless dielectrics

$$\begin{aligned} \omega &= 10^8 \\ \beta &= 1 \end{aligned}$$

$$v_p = \frac{3 \times 10^8}{\sqrt{\epsilon_R}} = 10^8$$

$$v_p = \frac{\omega}{\beta} = 10^8 \neq 3 \times 10^8$$

$$\Rightarrow \epsilon_R = 9$$

Propagation direction $\rightarrow a_y$

$$f = \frac{10^8}{2\pi}$$

$$\lambda = \frac{1}{\cancel{f}} 2\pi$$

$$H(y, t) = \left(\frac{25}{\frac{120\pi}{\sqrt{9}}} \right) \sin(10^8 t - y) (a_y \times a_z)$$

$$= \left(\frac{5}{8\pi} \right) \sin(10^8 t - y) a_x$$

1. $H = (0.5 e^{-0.1x}) \cos(10^6 t - 2x) a_z$

$$\omega = 10^6 \text{ rad/second}$$

$$\beta = 2 = \frac{2\pi}{\lambda} \Rightarrow \lambda = 3.14$$

Propagation $\rightarrow +a_x$

$\rightarrow$ Polarization means E field direction.

Note:-

Polarization direction is always E field orientation and direction

2. Free Space

$$E = 50 \sin(10^7 t + \beta z) a_y$$

$$\rightarrow \beta = \frac{10^7}{3 \times 10^8} = \frac{1}{30} = \frac{2\pi}{\lambda}$$

$$\Rightarrow \lambda = 60\pi = 188\text{m}$$

$$\rightarrow k = \beta = \frac{1}{30} = 0.033$$

$\rightarrow$ Free space, no attenuation

$$H = 0.5 e^{-j0.15z} \sin(10^6 t - 2x) a_z$$

$$v_p = \frac{10^6}{2} = 5 \times 10^5$$

$$V = 0.1 + j2$$

$$\text{direction} \rightarrow +a_x$$

$$\epsilon_r = 81$$

$$\omega = 6\pi \times 10^8, \quad \beta = ?$$

$$V = \frac{3 \times 10^8}{\sqrt{81}} = \frac{6\pi \times 10^8}{\beta}$$

$$\Rightarrow \beta = 18\pi$$

$$\mu_r = 2, \quad \epsilon_r = 8$$

$$\eta = \sqrt{\frac{\mu_0 \mu_r}{\epsilon_0 \epsilon_r}} = 120\pi \sqrt{\frac{2}{8}} = 188 \Omega$$

$$H = 0.1 \cos(\omega t - \beta z) a_x$$

$$E = (0.1 \times 377) \cos(\omega t - \beta z) (a_x \times a_z)$$

$$= -37.7 \cos(\omega t - \beta z) a_y$$

Ans - D

$$\beta = \omega \sqrt{\mu_0 \epsilon_0 \epsilon_r} = \frac{2\pi}{\lambda}$$

$$\lambda \propto \frac{1}{\sqrt{\epsilon}}$$

$$\frac{\lambda_0}{\lambda} = \sqrt{\frac{\epsilon_0 \epsilon_r}{\epsilon_0}} = \frac{2}{1} \Rightarrow \epsilon_r = 4$$

$$E = 20 e^{-j5z} a_x - 25 e^{-j5z} a_y$$

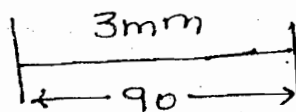
$$H = \left(\frac{20}{120\pi} \right) e^{-j5z} a_y - 25 e^{-j5z} (-a_x)$$

$$= \left(\frac{25}{120\pi} \right) e^{-j5z} a_x + \left(\frac{20}{120\pi} \right) e^{-j5z} a_y$$

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$$\beta = \frac{\theta}{z} = \frac{2\pi}{\lambda}$$

$$= \frac{\pi/2}{3\text{mm}} = \frac{2\pi}{\lambda}$$



$$\Rightarrow \lambda = 12\text{mm}$$

$$\lambda f = 12 \times 10^{-3} \times 10 \times 10^9 = \frac{3 \times 10^8}{\sqrt{\epsilon_R}}$$

$$\Rightarrow \epsilon_R = 6.25$$

20

$$E = A \cos\left(\omega t - \frac{\omega}{c} z\right) a_y$$

$$\frac{\omega}{c} = \beta$$

$$H = \left[\frac{A}{\sqrt{\mu/\epsilon}} \right] \cos\left(\omega t - \frac{\omega}{c} z\right) (-a_x)$$

$$= -A \sqrt{\frac{\epsilon}{\mu}} \cos\left(\omega t - \frac{\omega}{c} z\right) a_x$$

Ans-19

21

$$\nabla^2 E + k^2 E = 0$$

$$e^{-\gamma z} + e^{\gamma z}$$

$$\nabla^2 E = \gamma^2 E$$

$$\nabla^2 m^2 = 0$$

$$\nabla^2 E - \gamma^2 E = 0$$

$$\text{if } \gamma = j\beta$$

$$e^{-j\beta z} + e^{j\beta z}$$

$$\nabla^2 E + \beta^2 E = 0$$

$$(\nabla^2 + m^2 = 0)$$

$$(z)$$

Ans-C

22

$$E(x, z, t)_y = 25 \sin(\omega t - 3x + 4z) a_y$$

$$\text{Unit Prop. direction} = \frac{3a_x - 4a_z}{5}$$

$$\beta = \sqrt{\beta_x^2 + \beta_z^2} = 5 = \frac{2\pi}{\lambda}$$

Note -

$$\nabla^2 E = \mu \epsilon \frac{\partial^2 E}{\partial t^2}$$

$$\frac{\partial^2 E}{\partial z^2} = \mu \epsilon \frac{\partial^2 E}{\partial t^2} = 0$$

$$E(z, t) = E_0 e^{j\omega t} \cdot e^{-j\beta z}$$

$$\frac{\partial^2 E}{\partial x^2} + \frac{\partial^2 E}{\partial z^2} - \mu \epsilon \frac{\partial^2 E}{\partial t^2} = 0$$

$$E(x, z, t) = E_0 e^{j\omega t} \cdot e^{-j\beta_x x} \cdot e^{-j\beta_z z}$$

It is a product solution of time, x & z harmonics

$$E(x, z, t) = 25 \sin(\omega t - 3x + 4z) a_y$$

$$H(x, z, t) = \left(\frac{25}{120\pi} \right) \sin(\omega t - 3x + 4z)$$

$$= \frac{(3a_x - 4a_z)}{5} \times a_y$$

$$= \left(\frac{25}{120\pi} \right) \sin(\omega t - 3x + 4z) \left(\frac{3a_z + 4a_x}{5} \right)$$

$$E(x, z, t)_y$$

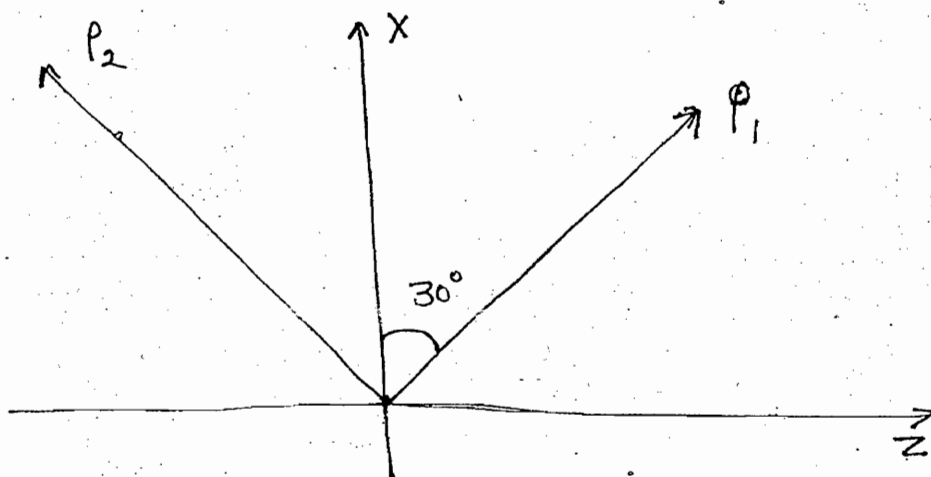
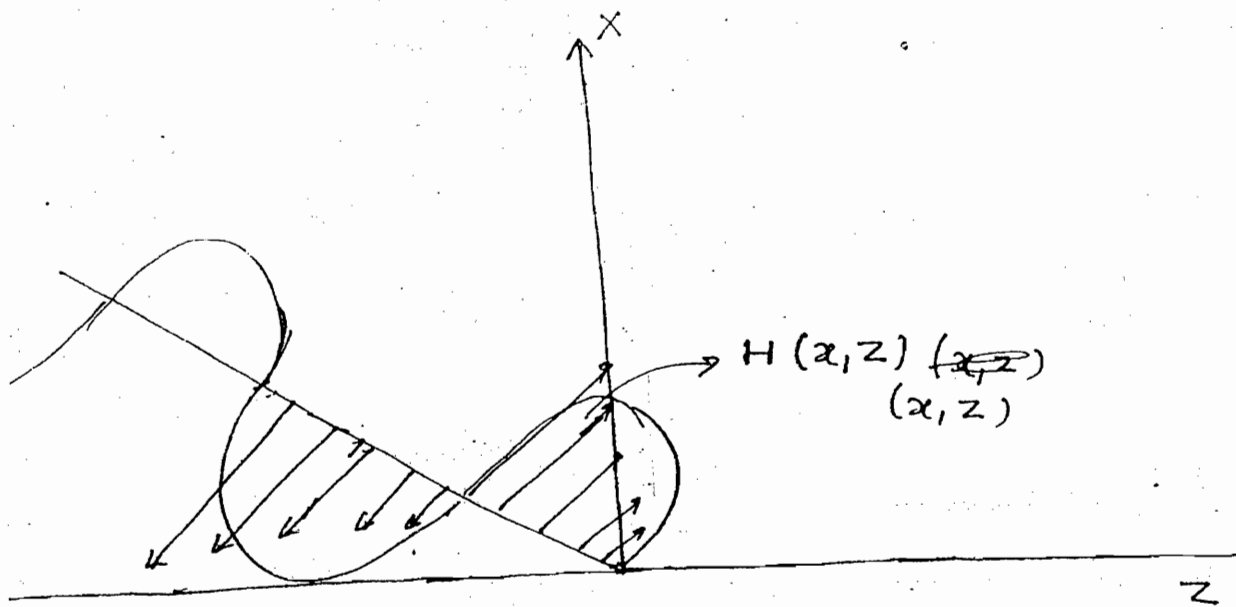
$$H(x, z, t)(x, z)$$

$$\text{Prop. direction} \rightarrow 3a_x - 4a_z$$

$$H \text{ direction} \rightarrow 4a_x + 3a_z$$

$$E \text{ direction} \rightarrow a_y \quad (\because \text{dot product} = 0)$$

$$\boxed{E \perp H \perp P} \rightarrow \text{Basic Transverse Nature}$$



90° to Y -axis $\Rightarrow$ zx Plane

30° to the x -axis

$$E(x, z, t) = E_0 e^{j\omega t} e^{-j\beta_x x} e^{\pm j\beta_z z}$$

$$\tan 30^\circ = \frac{\beta_z}{\beta_x} = \frac{1}{\sqrt{3}}$$

$$\beta_x = \sqrt{3} \beta_z$$

$$\beta = \frac{2\pi}{\lambda} = \sqrt{\beta_x^2 + \beta_z^2} = \sqrt{(\sqrt{3} \beta_z)^2 + \beta_z^2}$$

Ans - (a)

$$\Rightarrow \beta_z = \frac{\pi}{\lambda} \quad \beta_x = \frac{\sqrt{3}\pi}{\lambda}$$

$$\beta = 280\pi = \frac{2\pi}{\lambda}$$

$$\lambda = \frac{1}{140}$$

$$f = 14 \times 10^9$$

$$\lambda f = 10^8 = \frac{3 \times 10^8}{\sqrt{\epsilon_R}} \Rightarrow \epsilon_R = 9$$

$$\frac{E_p}{3} = \frac{120\pi}{\sqrt{9}} \Rightarrow E_p = 120\pi, \text{ Ans}$$

Lecture - 10

Note:-

$R, L, G, C \rightarrow$ Primary constant $-\frac{V}{I}$ Waves
|

Transmission
line

$\mu, \sigma, \epsilon \rightarrow$ Material constant $-\frac{E}{H}$ Waves
|

Materials

$$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)} \quad Y = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)}$$
$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}} \quad \eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}}$$

Note:- (1)

$$\mu = \frac{|\gamma\eta|}{\omega}$$

$$\sigma = \text{Real} \left[\frac{\gamma}{\eta} \right]$$

$$\epsilon = \frac{\text{Imag} \left[\frac{\gamma}{\eta} \right]}{\omega}$$

Note 2:-

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}} = \text{Complex} = \frac{E_x}{H_y} = |\eta| \angle \theta$$
$$= \left| \frac{E_x}{H_y} \right| \angle \theta$$

- In general material E_x and H_y are orthogonal in directions but have a phase shift or time delay which is η 's phase

Example :-

$$E \rightarrow \sin(\omega t - \beta z) a_x$$

$$H \rightarrow \sin(\omega t - \beta z - \theta) a_y$$

Note 3:-

$$\gamma = \alpha + j\beta = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)}$$

$$\alpha^2 - \beta^2 + j2\alpha\beta = j\omega\mu\sigma - \omega^2\mu\epsilon$$

$$\alpha = \omega \sqrt{\frac{\mu\epsilon}{2} \left(\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2} - 1 \right)}$$

$$\beta = \omega \sqrt{\frac{\mu\epsilon}{2} \left(\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2} + 1 \right)}$$

Loss Tangent or Dissipation Factor:-

- $\frac{\sigma}{\omega\epsilon}$ decides the propagation aspects α or β of the EM waves

If $\frac{\sigma}{\omega\epsilon} \gg 1 \Rightarrow$ very good conductor

α is very large $\Rightarrow$ EM Wave propagation is very tough

- EM waves cannot easily propagate in good conductors

- The propagation aspects depends on $\frac{\sigma}{\omega\epsilon}$ and hence on frequency - ω also

eg:- $\frac{\sigma}{\omega\epsilon} \gg 1$ For human body upto light frequency

$\frac{\sigma}{\omega\epsilon} \ll 1$ For human body beyond X-ray frequency

eg:- $\frac{\sigma}{\omega\epsilon} \gg 1$ For earth upto MHz only

$\frac{\sigma}{\omega\epsilon} \ll 1$ for earth beyond microwaves

summary:-

The term $\frac{\sigma}{\omega\epsilon} = \left| \frac{J_c}{J_d} \right|$ = loss tangent and hence decides the behaviour of the material

$J_c \gg J_d \Rightarrow$ good conductor

$J_d \gg J_c \Rightarrow$ good dielectric

$$\frac{J_c}{J_d} = \frac{\sigma E}{\epsilon \frac{\partial E}{\partial t}} = \frac{\sigma E}{j\omega\epsilon E} = \frac{\sigma}{\omega\epsilon}$$

Case - (III) :-

EM Wave Propagation in very good conductors
($\sigma \gg \omega\epsilon$)

$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)} \simeq \sqrt{j\omega\mu\sigma}$$

$$= \sqrt{\omega\mu\sigma} \angle 90^\circ = \sqrt{\omega\mu\sigma} \angle 45^\circ$$

$$= \sqrt{\frac{\omega\mu\sigma}{2}} + j\sqrt{\frac{\omega\mu\sigma}{2}}$$

$$\alpha = \beta = \sqrt{\frac{\omega\mu\sigma}{2}}$$

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}} = \sqrt{\frac{j\omega\mu}{\sigma}} = \sqrt{\frac{\omega\mu}{\sigma}} \angle 90^\circ$$

$$= \sqrt{\frac{\omega\mu}{\sigma}} \angle 45^\circ = \sqrt{\frac{\omega\mu}{2\sigma}} + j\sqrt{\frac{\omega\mu}{2\sigma}}$$

$$R = X = \sqrt{\frac{\omega\mu}{2\sigma}}$$

Note (1): -

→ η 's real part (R) = Cause of attenuation (α)

η 's Imaginary part (X) = Cause of phase shift (β)

η 's phase = 45° means E_x & H_y are delayed in time by 45°

Skin depth or Depth of penetration: -

$$\begin{array}{c} |E| = E_0 e^{-\alpha z} \\ E_0 = 100 \\ \downarrow \quad \quad \quad 1/e \\ z = \frac{1}{\alpha} \quad \quad \quad 37 \\ \downarrow \quad \quad \quad 1/e^2 \\ z = \frac{2}{\alpha} \quad \quad \quad 13 \\ \downarrow \quad \quad \quad 1/e^3 \\ z = \frac{3}{\alpha} \quad \quad \quad 4 \end{array}$$

→ The first $\frac{1}{\alpha}$ distance travelled by the wave is where majority of the wave amplitude is attenuated. This distance is called as skin depth. Hence skin depth is the distance travelled by the wave where the wave leakage to $1/e$ times (37%) of the initial value.

$$\delta = \frac{1}{\alpha} = \sqrt{\frac{2}{\omega \mu \sigma}} = \sqrt{\frac{2}{\omega \mu}} \cdot \frac{1}{\sigma} = \frac{1}{\sigma R_s}$$

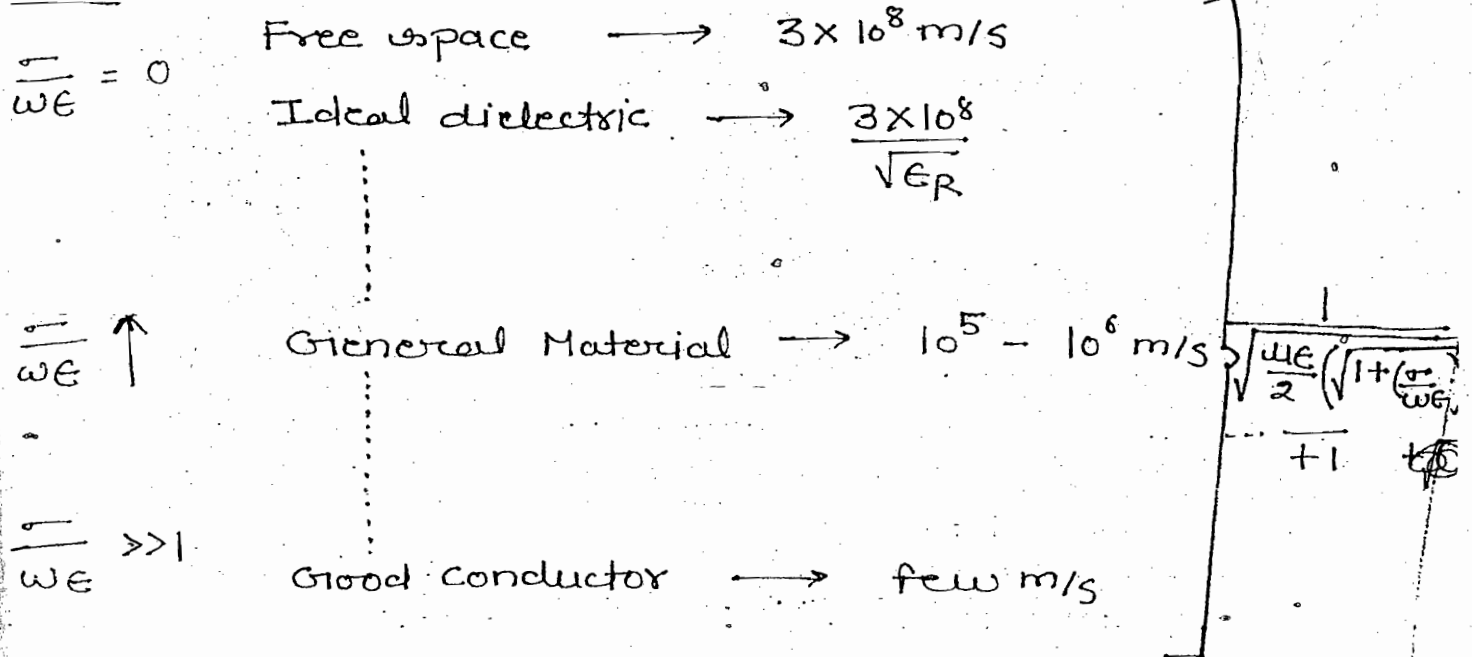
where $R_s = \eta$'s real part = skin resistance

15. $V_p = \frac{\omega}{\beta} = \frac{\omega}{\omega \delta} = \omega \delta = 2\pi \times 2 \times 10^5 \times 4 \times 10^{-6}$
 $= 1.6\pi \text{ m/s} \approx 5 \text{ m/s}$

16. $\omega' = 4\omega$ $\delta = \sqrt{\frac{2}{\omega \mu \sigma}} \Rightarrow \delta' = \frac{\delta}{2}$

$V_p' = \omega' \delta' = 4\omega \frac{\delta}{2} = 2V_p = 10 \text{ m/s}$

Note:-



$$V_p = \frac{\omega}{\mu \sqrt{\frac{\mu \epsilon}{2} \left(\sqrt{1 + \left(\frac{\sigma}{\omega \epsilon} \right)^2} + 1 \right)}}$$

17. $\frac{\sigma}{\omega \epsilon} = 1.73$

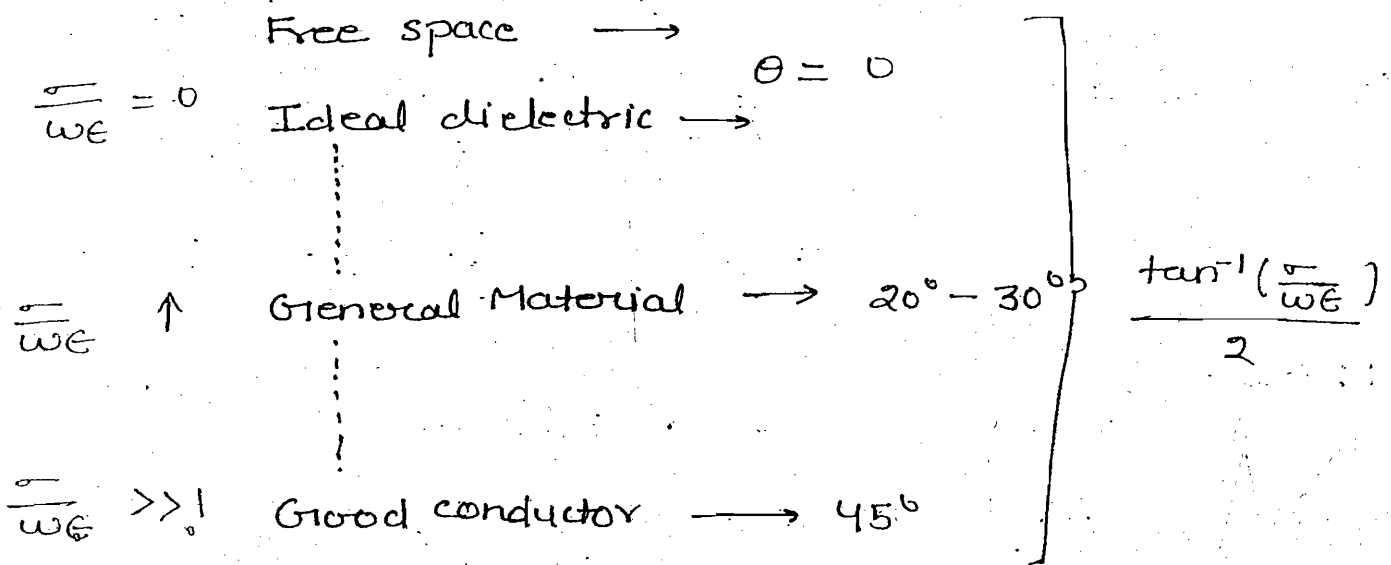
η 's phase $= \sqrt{\frac{j\omega \mu}{\sigma + j\omega \epsilon}} = \sqrt{\frac{\omega \mu \angle 90}{K \angle \tan^{-1}(\frac{\omega \epsilon}{\sigma})}}$

$= \sqrt{K \angle 90 - \tan^{-1}(\frac{\omega \epsilon}{\sigma})}$ $\frac{\tan^{-1}(\sigma/\omega \epsilon)}{2}$

η 's phase $= \frac{90 - \tan^{-1}(\frac{\omega \epsilon}{\sigma})}{2} = \frac{\tan^{-1} \sqrt{3}}{2} = \frac{60}{2} = 30^\circ$

Note:-

η 's phase = E_x to H_y Phase



28.

$$J_c = J_d$$

$$\frac{\sigma}{\omega\epsilon} = 1$$

$$\Rightarrow \frac{\sigma}{2\pi f\epsilon} = 1$$

$$f = \frac{10^{-2}}{2\pi \times 4 \times 1} = 45 \text{ MHz}$$

$$36\pi \times 10^9$$

29.

$$\frac{I_c}{I_d} = \frac{J_c}{J_d} = \frac{\sigma}{\omega\epsilon}$$

$$\Rightarrow I_d = \frac{I_c \omega \epsilon}{\sigma} = \frac{1 \times 2\pi \times 50 \times 8.8 \times 10^{-12}}{58}$$

Ans - (B)

30.

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma}} = \sqrt{\frac{\omega\mu}{\sigma}} \angle 45^\circ$$

$$= R + jX$$

Ans - D

31.

$$\eta = 0.02 \angle 45^\circ$$

Ans - (a)

32.

lossy dielectric, $\frac{\sigma}{\omega\epsilon} \ll 1$

$$\begin{aligned} Y &= \sqrt{j\omega\mu(1 + j\frac{\sigma}{\omega\epsilon})} = \sqrt{j\omega\mu - j\omega\epsilon(1 + \frac{\sigma}{j\omega\epsilon})} \\ &= j\omega\sqrt{\mu\epsilon} \left(1 + \frac{\sigma}{j2\omega\epsilon}\right) = \frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon}} + j\omega\sqrt{\mu\epsilon} \end{aligned}$$

Power, Power density and Poynting Vector :-

→ Energy that can be transformed into other formats is said to be a power format

$$\frac{\text{Joules}}{\text{second}} = \text{Watts}$$

Using Maxwell's IV Equation

$$\nabla \times H = \sigma E + \epsilon \frac{\partial E}{\partial t}$$

Take $\cdot E$ on both sides

$$(\nabla \times H) \cdot E = \sigma E^2 + \epsilon \frac{\partial E}{\partial t} \cdot E$$

$$\Rightarrow -\nabla \cdot (E \times H) + H \cdot (\nabla \times E) = \sigma E^2 + \epsilon \frac{\partial E}{\partial t} \cdot E$$

$$\Rightarrow -\nabla \cdot (E \times H) - \sigma E^2 = \epsilon \frac{\partial E}{\partial t} \cdot E + \mu \frac{\partial H}{\partial t} \cdot H$$

$$= -\frac{\partial}{\partial t} \left(\frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2 \right)$$

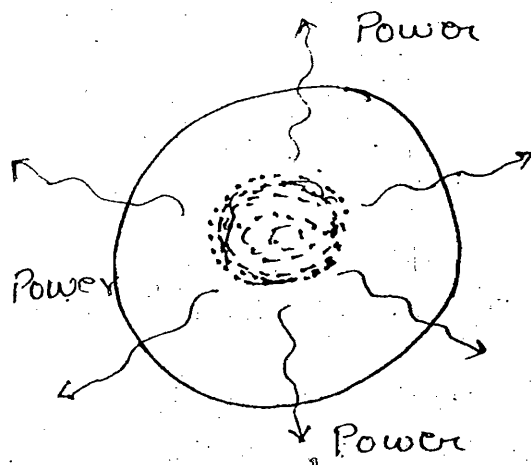
Take $\int dv$ on both sides

$$\int \nabla \cdot (E \times H) dv + \int \sigma E^2 dv = -\frac{\partial}{\partial t} \int \left(\frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2 \right) dv$$

Using Divergence theorem

$$\oint (E \times H) \cdot d\mathbf{s} + \int \epsilon E^2 dV = \text{Rate of decrease of } E/H$$

Energy inside a volume



The LHS is a EM wave power crossing the surface

Hence $E \times H$ = Power density of EM Wave
 = strength of power of EM Wave
 = Poynting Vector of the EM Wave
 = It has the direction of Propagation

$$E(z,t)_x \times H(z,t)_y = P(z,t)_z$$

= Power density at a time at a point in space

= Instantaneous Poynting Vector

$P(z)_{avg}$ = Time Average Poynting Vector

$$= E_{RMS} \cdot H_{RMS} = \frac{E_0 e^{-\alpha z}}{\sqrt{2}} \cdot \frac{H_0 e^{-\alpha z}}{\sqrt{2}} \therefore$$

$$P(z)_{avg} = \frac{1}{2} E_0 H_0 e^{-2\alpha z} = \frac{1}{2} \frac{E_0^2}{\eta} e^{-2\alpha z} = \frac{1}{2} \eta H_0^2 e^{-2\alpha z}$$

Note 1:-

The average power decays at 2x rate exponentially in the medium

→ This power lost in the wave is acquired by the medium as ohmic power

$$J \cdot E \text{ or } -E^2 \text{ Watts/m}^3$$

Note 2:-

Alternatively

$$P(z)_{avg} = \frac{1}{2} (E \times H^*)$$

* conjugate operation means average power is independent of harmonic and its phase and depends only on amplitude.

> In free space

$$P_{avg} = \frac{1}{2} E_0 H_0 = \frac{1}{2} \frac{E_0^2}{\eta} = \frac{1}{2} \eta H_0^2$$

Workbook:-

33. $P_{avg} = \frac{1}{2} \eta H_0^2 (-a_y)$

$$H_0 = 0.1$$

$$= \frac{1}{2} \cdot 120\pi \cdot (0.1)^2 (-a_y) \rightarrow \underline{B}$$

34. $W = \int P_{avg} \cdot ds$

$$= \int \frac{1}{2} \frac{E_0^2}{\eta} a_x \cdot ds a_x \quad \nearrow x = \text{constant}$$

$$= \frac{1}{2} \cdot \frac{60 \cdot 60}{120\pi} \pi \times 4^2 = 240W$$

35. C

36. (1) $H = 50 \sin(\omega t - \beta z) a_x + 150 \sin(\omega t - \beta z) a_y$

$$P_{avg} = \frac{1}{2} \eta H_0^2 (a_z)$$

$$H = 50 \sin(\omega t - \beta z) \left(\frac{a_x + 3a_y}{\sqrt{10}} \right) \sqrt{10}$$

$$= (50\sqrt{10}) \sin(\omega t - \beta z) \left(\frac{a_x + 3a_y}{\sqrt{10}} \right)$$

$$P_{avg} = \frac{1}{2} \cdot 120\pi \cdot (50\sqrt{10})^2 a_z$$

$$H_0 = 50\sqrt{10}$$

$$\begin{aligned} \text{(ii)} \quad H &= 50 \sin(\omega t - \beta z) a_x + 150 \cos(\omega t - \beta z) a_y \\ &= 50 \sin(\omega t - \beta z) a_x (1 + 3j) \\ &= 50 (1 + 3j) \sin(\omega t - \beta z) a_x \end{aligned}$$

Note:-

Orthogonal field components in time ^{or} ~~and~~ space have same effect

$$\text{(ii)} \quad H = 50 \sin(\omega t - \beta z) a_x + 150 \cos(\omega t - \beta z) a_y$$

$$H = 50 \sin(\omega t - \beta z) (a_x + 3j a_y)$$

$$P_{avg} = \frac{1}{2} (E \times H^*)$$

$$E = (50\eta) \sin(\omega t - \beta z) (-a_y) + (150\eta) \cos(\omega t - \beta z) (a_x)$$

$$= (50\eta) \sin(\omega t - \beta z) (3j a_x - a_y)$$

$$P_{avg} = \frac{1}{2} \cdot 50 \cdot 120\pi (3j a_x - a_y) \times 50 (a_x - 3j a_y)$$

$$= \frac{1}{2} \cdot 120\pi \cdot 50^2 (a_z + 9a_z)$$

$$= \frac{1}{2} \cdot 120\pi (50\sqrt{10})^2 a_z$$

$$\underline{37.} \quad E = (a_x + ja_y) e^{jkz - j\omega t}$$

$$H = \frac{k}{\omega\mu} (a_y - ja_x) e^{jkz - j\omega t}$$

$$P_{avg} = \frac{1}{2} (a_x + ja_y) \times \frac{k}{\omega\mu} (a_y - ja_x)$$

$$= \frac{k}{2\omega\mu} (a_z - a_z) = 0 \rightarrow \text{Null vector}$$

Conventional :-

$$\underline{1.} \quad E = 100 (ja_x + 2a_y - ja_z) e^{j\omega t}$$

$$H = (-a_x + ja_y + a_z) e^{j\omega t}$$

$$P_{avg} = \frac{1}{2} (E \times H^*) = \frac{1}{2} 100 (ja_x + 2a_y - ja_z) \times (-a_x - ja_y + a_z)$$

$$= 50 \begin{vmatrix} a_x & a_y & -a_z \\ j & 2 & -j \\ -1 & -j & 1 \end{vmatrix}$$

$$= 50 (3a_x + 3a_z)$$

$$|P_{avg}| = 150\sqrt{2}$$

$$P_{avg} \text{ direction} = (a_x + a_z)$$

$$\text{Unit } P_{avg} \text{ direction} = \frac{a_x + a_z}{\sqrt{2}}$$

Wave Polarization:-

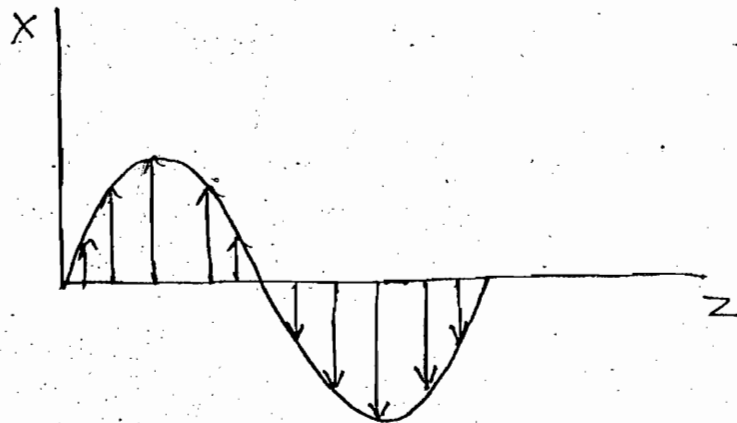
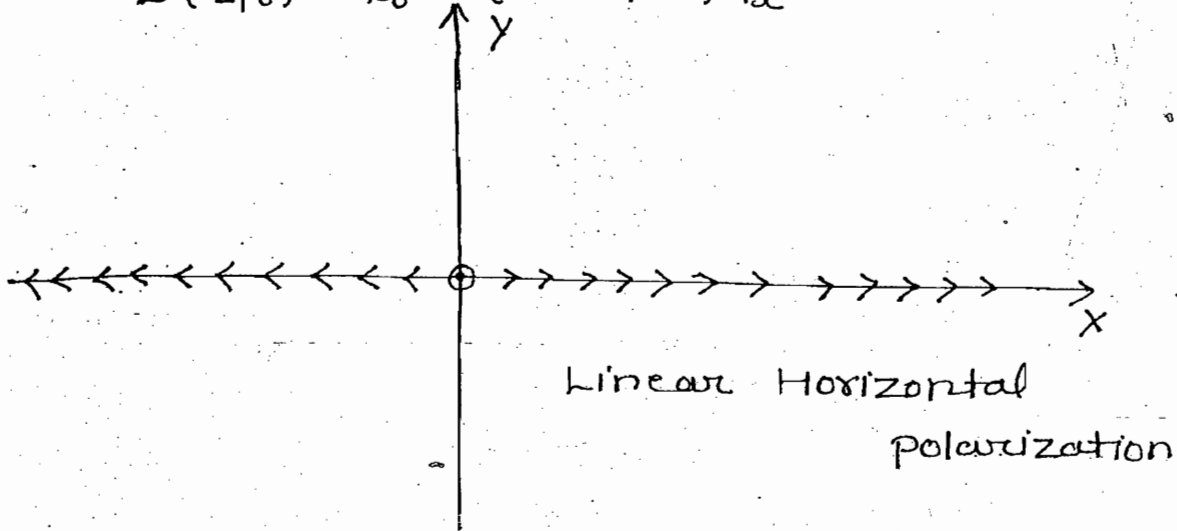
It is the electric field orientation & the possible planar components of the E field satisfying the basic transverse nature

EM Wave is propagating along z-axis in the following cases:-

Case (I):-

E field is oriented in x-direction

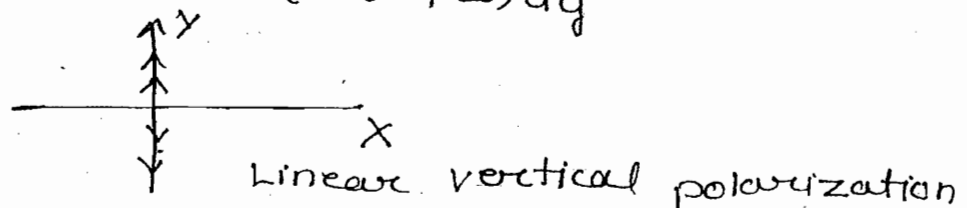
$$E(z,t) = E_0 \sin(\omega t - \beta z) a_x$$



Case - (II):-

E field is oriented in y-direction

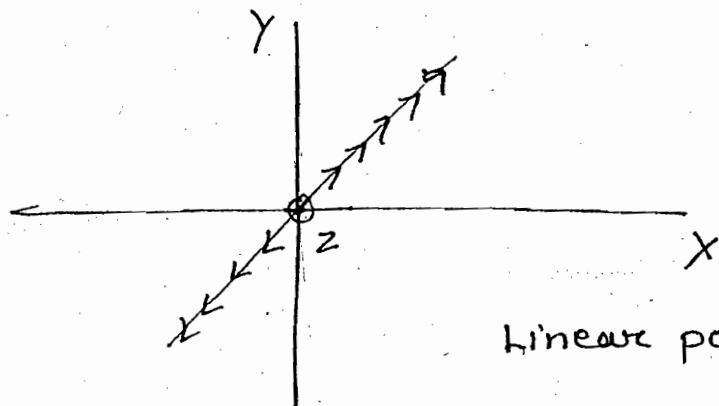
$$E(z,t) = E_0 \sin(\omega t - \beta z) a_y$$



Case-(III) :-

$\vec{E}$ field is oriented in x & y directions

$$\vec{E}(z,t) = E_1 \sin(\omega t - \beta z) \hat{a}_x + E_2 \sin(\omega t - \beta z) \hat{a}_y$$



Linear polarization

Note! -

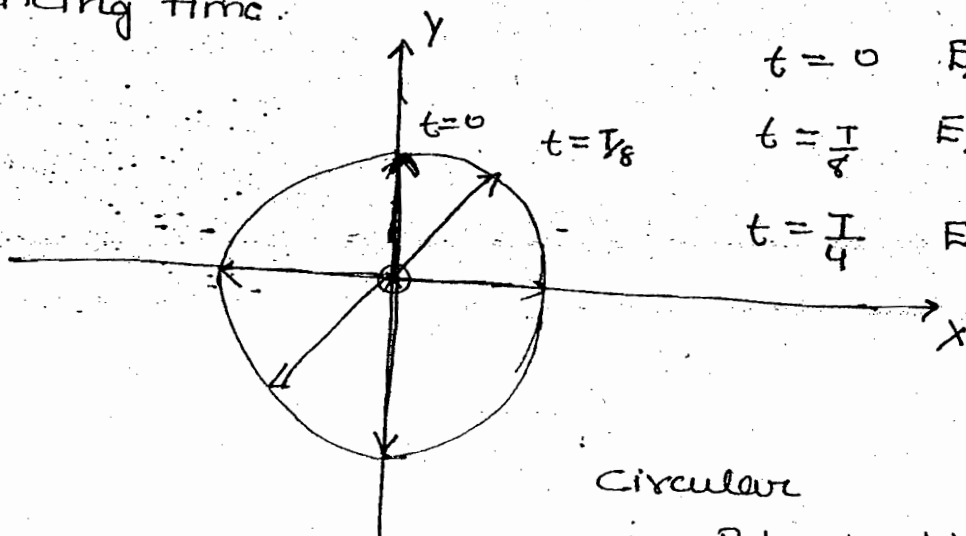
If the EM waves has single $\vec{E}$ field components or two planar components both inphase the wave is said to be linearly polarized

Case-(IV) :-

$\vec{E}$ field is oriented in x & y directions

$$\vec{E}(z,t) = E_0 \sin(\omega t - \beta z) \hat{a}_x + E_0 \cos(\omega t - \beta z) \hat{a}_y$$

→ Polarization is identified by studying the trace of the $\vec{E}$ field on the $z=0$ plane for various advancing time.



Circular

Polarization.

41

Summary:-

If the EM Waves has two planar components of E field both out of phase at 90° and equal amplitude, the wave is circularly polarised.

Note:-

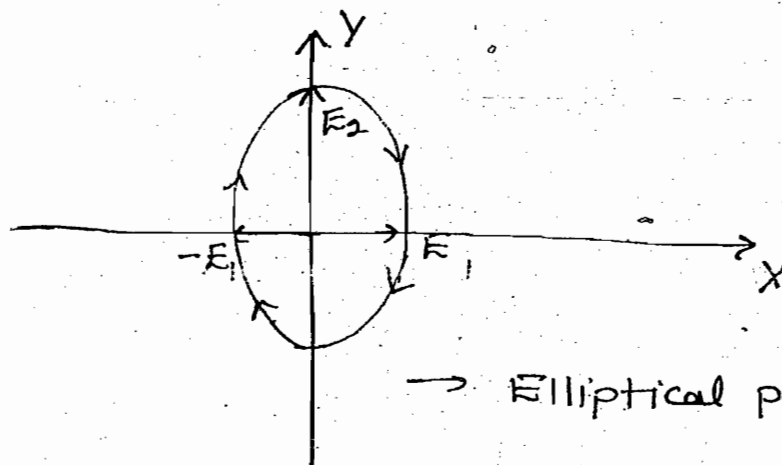
Sense of rotation left or right, if the left hand thumb points towards propagation direction and the closed finger along advancing time, the wave is left circularly polarised

eg:- out of paper - propagation clockwise time advancement - left circular

Case-(V):-

E field is oriented in x & y directions

$$E(z,t) = E_1 \sin(\omega t - \beta z) a_x + E_2 \cos(\omega t - \beta z) a_y$$

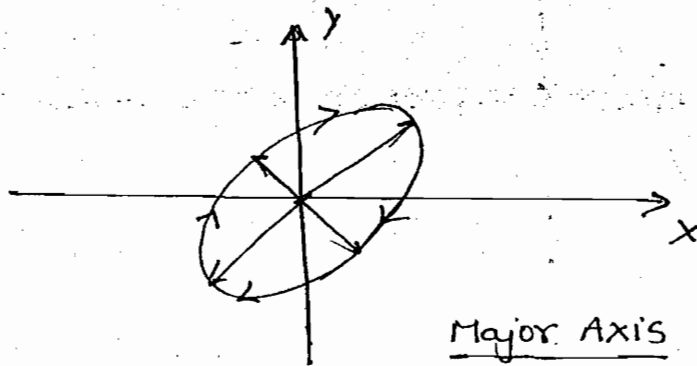


→ Elliptical polarization

Case-(VI):-

E field is oriented in x & y direction

$$E(z,t) = E_1 \sin(\omega t - \beta z) a_x + E_2 \sin(\omega t - \beta z + \theta) a_y$$



Major Axis = $AR = [1, \infty)$

Minor Axis

Circle

Linear

Workbook :-

i) $\vec{E} = 25 \sin(\omega t + 4x) (a_y + 6a_z)$

→ Linear

ii) $\vec{E} = 25 \sin(\omega t + 4x) a_y + 25 \cos(\omega t + 4x) a_z$

→ Circular

iii) $\vec{E} = 25 \sin(\omega t + 4x + 60^\circ) a_y + 25 \cos(\omega t + 4x + 60^\circ) a_z$

→ Circular

iv) $\vec{E} = 25 \sin(\omega t + 4x) a_y + (25\sqrt{2}) \sin(\omega t + 4x + 45^\circ) a_z$

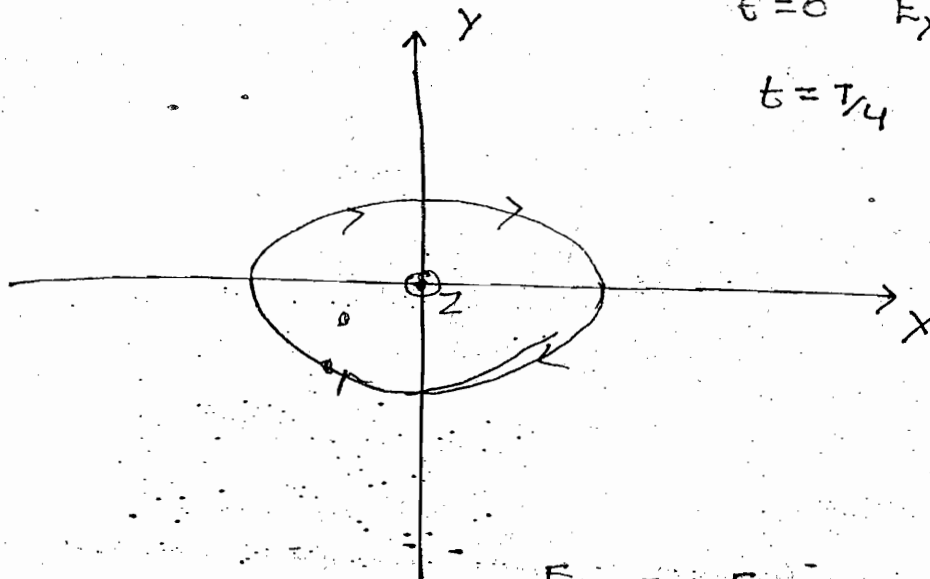
→ Elliptical

v) Elliptical

vi) Circular

g. $\theta = 60^\circ \rightarrow$ elliptical

h. $\theta = 90^\circ$



$t=0 \quad E_x = E_1 \neq E_y = 0$

$t=T/4 \quad E_x = 0 \quad E_y = E_2$

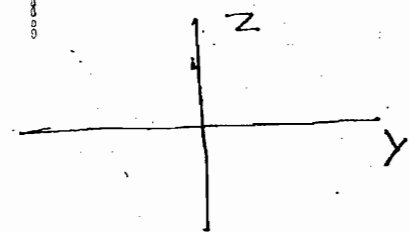
Summary:-

E_x	E_y	Prop.	Polarization
$\cos$	$-\sin$	z	Left
$\cos$	$\sin$	z	Right
$\cos$	$-\sin$	$-z$	Right
$-\sin$	$\cos$	z	Right

$\left. \begin{array}{l} E_x \rightarrow E_1 \cos \omega t \\ E_y \rightarrow E_2 \sin \omega t \end{array} \right\} e^{jKz}$

$(ay + jaz) e^{j\omega t}$
 $\downarrow$
 $\sin$
 $\downarrow$
 $\cos$

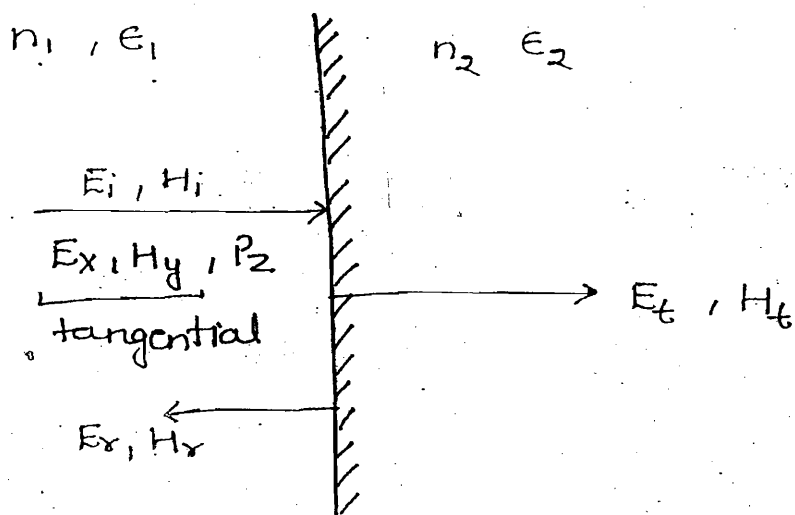
$\downarrow$
 $\cos$
 $\downarrow$
 $-\sin$



Lecture - 11

Reflections / Transmissions of EM Waves.

- (I) Normal Incidence (II) Dielectric - Dielectric
 $z < 0$ $z = 0$ (XY Plane)



- (I) $E_t > E_i$ or $E_t < E_i \rightarrow$ It depends on ϵ_1 & ϵ_2
 (II) $H_t > H_i$ if $E_t < E_i$
 (III) $H_t < H_i$ if $E_t > E_i$
 (IV) $P_t < P_i \rightarrow$ Always

E_x	H_y	Prop. z	
a_x	a_y	a_z	$\rightarrow$ Incidence
a_x	$-a_y$	$-a_z$	$\left. \begin{array}{l} \rightarrow \\ \rightarrow \end{array} \right\}$ Reflection
$-a_x$	a_y	$-a_z$	

$\rightarrow$ When reflections occur either E or H negates along with propagation direction but not both

$$E_i = n_1 H_i$$

$$E_t = n_2 H_t$$

$$E_r = -n_1 H_r$$

$\rightarrow$ For normal incidence the E/H fields are tangential.

$$E_{t1} = E_{t2} \rightarrow E_i + E_r = E_t$$

$$H_{t1} = H_{t2}$$

$$E_i + E_r = E_t$$

$$1 + \frac{E_r}{E_i} = \frac{E_t}{E_i}$$

$$\Rightarrow \boxed{1 + \Gamma = \tau} \quad \text{--- (1)}$$

$$\text{where } \Gamma = \frac{E_r}{E_i}$$

= Reflection coefficient
for the E fields

$$\tau = \frac{E_t}{E_i} = \text{Transmission coefficient
for the E fields}$$

$$H_i + H_r = H_t$$

$$\frac{E_i}{n_1} - \frac{E_r}{n_1} = \frac{E_t}{n_2}$$

$$E_i - E_r = \frac{n_1}{n_2} E_t$$

$$\boxed{\Gamma = \frac{n_2 - n_1}{n_2 + n_1}}$$

Note :-

$$\Gamma = \frac{n_1 - n_2}{n_1 + n_2} = -\Gamma_E$$

$$\Gamma_P = \Gamma_E \cdot \Gamma_H = -\Gamma_E^2 = -\Gamma_H^2$$

$$|1 + \Gamma| = \tau$$

Always - Either E or H or Power

(iv) If $n_1 = \sqrt{\frac{\mu_0}{\epsilon_1}}$ & $n_2 = \sqrt{\frac{\mu_0}{\epsilon_2}}$

$$\Gamma_E = \frac{\sqrt{\epsilon_1} - \sqrt{\epsilon_2}}{\sqrt{\epsilon_1} + \sqrt{\epsilon_2}}$$

If $\epsilon_1 > \epsilon_2 \Rightarrow \Gamma_E = +ve \quad \tau_E > 1$

$$E_t > E_i$$

$$\Rightarrow \Gamma_H = -ve, \quad \tau_H < 1$$

$$H_t < H_i$$

$$\Gamma_P = -ve, \quad \tau_P < 1 \rightarrow \text{Always}$$

Workbook:-

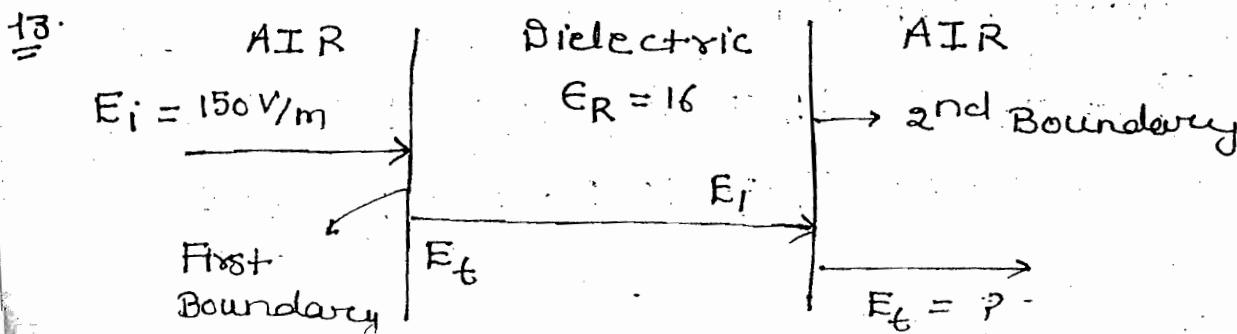
12. $\frac{\text{Air}}{\epsilon_0} - \frac{\text{Dielectric}}{4\epsilon_0}$

$$\tau_P = ?$$

$$\Gamma_E = \frac{\sqrt{\epsilon_0} - \sqrt{4\epsilon_0}}{\sqrt{\epsilon_0} + \sqrt{4\epsilon_0}} = \frac{1-2}{1+2} = -\frac{1}{3}$$

$$\Gamma_H = +\frac{1}{3} \quad \Gamma_P = -\frac{1}{9}$$

$$\Rightarrow \tau_P = 1 - \frac{1}{9} = \frac{8}{9}$$



At the 1st boundary

$$\Gamma_E = \frac{1 - \sqrt{16}}{1 + \sqrt{16}} = \frac{-3}{5}$$

$$\tau_E = 1 - \frac{-3}{5} = \frac{2}{5} = \frac{E_t}{E_i} \Rightarrow E_t = \frac{2}{5} \times 150 \text{ V/m} \\ = 60 \text{ V/m}$$

E_i at the 2nd Boundary = 60 V/m

$$\Gamma_E = \frac{\sqrt{16} - \sqrt{1}}{\sqrt{16} + \sqrt{1}} = \frac{3}{5}$$

$$\tau_E = 1 + \frac{3}{5} = \frac{8}{5} = \frac{E_t}{E_i}$$

$$E_t = \frac{8}{5} \times 60 = 96 \text{ V/m}$$

14. $\Gamma_E = \frac{1 - \sqrt{4}}{1 + \sqrt{4}} = \frac{-1}{3}$

$$\tau_E = 1 - \frac{-1}{3} = \frac{2}{3} = \frac{E_t}{E_i}$$

$$E_t = \frac{2}{3} E_0 \cos(\omega t - 2\beta z) a_y$$

$$(\beta' = \omega \sqrt{\mu_0 \epsilon_0 \epsilon_R} = 2\beta)$$

Note:-

$\alpha, \beta, V, \eta, V_p, \lambda \rightarrow$ Propagation aspects

& material aspects and change as ϵ, μ changes

$\omega \rightarrow$ source aspect / time aspect and is same for every material.

15. $\Gamma_E = -\frac{1}{3}$

$$H_i = \cos(10^8 t - \beta z) a_y$$

$$E_i = 120\pi \cdot \cos(10^8 t - \beta z) a_x$$

$$\frac{10^8}{\beta} = 3 \times 10^8 \Rightarrow \beta = \frac{1}{3}$$

$$E_y = \left(-\frac{1}{3} \cdot 120\pi\right) \cos\left(10^8 t + \frac{z}{3}\right) a_x$$

$$\frac{E_t}{E_y} = \frac{E_t}{E_i \frac{E_y}{E_i}} = \frac{\tau}{\Gamma} = \frac{1 + \Gamma}{\Gamma} = -2$$

$$\Gamma = -\frac{1}{3} = \frac{1 - \sqrt{\epsilon_2/\epsilon_1}}{1 + \sqrt{\epsilon_2/\epsilon_1}} \Rightarrow \frac{\epsilon_2}{\epsilon_1} = 4$$

48. EM Wave reflections at conductors:-

$$\Gamma_E = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

$$\eta_1 = 120\pi = 377$$

$$\eta_2 = \sqrt{\frac{j\omega\mu}{\sigma}} = 0$$

→ EM wave power is completely reflected at conductors

$$\Gamma_E = \frac{E_y}{E_i} = 1$$

$$E_i + E_y = E_t = 0$$

$$\tau_E = 0$$

$$\Gamma_H = \frac{H_y}{H_i} = 1$$

Ans-cc)

$$H_i + H_y = H_t = H_{max}$$

$$\tau_H = 2$$

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}} = \sqrt{\frac{\mu}{\epsilon}} \quad (\text{if } \sigma = 0) \text{ real}$$

$$= \sqrt{\frac{\mu^*}{\epsilon^*}} \quad (\text{if } \sigma \neq 0) \text{ complex}$$

$$\text{Ans } \mu^* = \epsilon^*$$

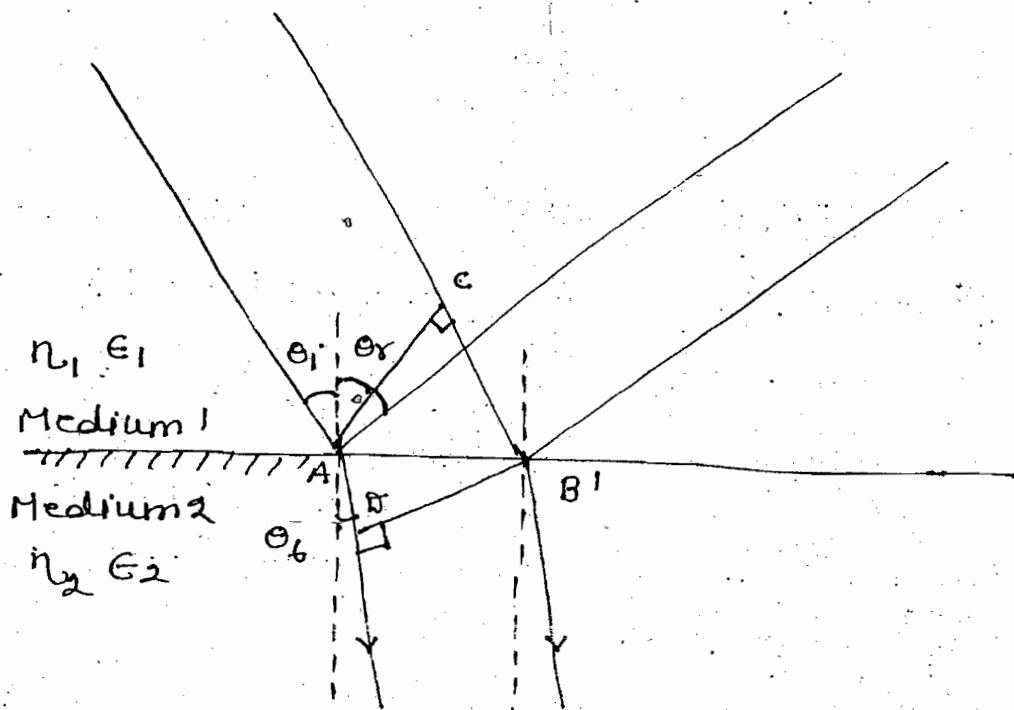
$$\eta = \sqrt{\frac{\mu^*}{\epsilon^*}} = 1$$

$$\Gamma = \frac{1 - 377}{1 + 377}$$

$$\Rightarrow \Gamma \approx -1$$

Ans - (c)

● Oblique (Inclined) Incidence :-



- A/C are inphase points in medium 1
- CB is the extra distance travelled in medium 1
- B/B are inphase points in medium 2
- AB is the extra distance travelled in medium 2

$$\frac{CB}{AB} = \frac{v_1}{v_2} = \text{Refracting Index of med. 2 w.r.t med 1}$$

$$\frac{AB \cos(90 - \theta_i)}{AB \cos(90 - \theta_t)} = \frac{1}{\sqrt{\mu_0 \epsilon_1}} \sqrt{\mu_0 \epsilon_2}$$

$$\boxed{\frac{\sin \theta_i}{\sin \theta_t} = \sqrt{\frac{\epsilon_2}{\epsilon_1}}}$$

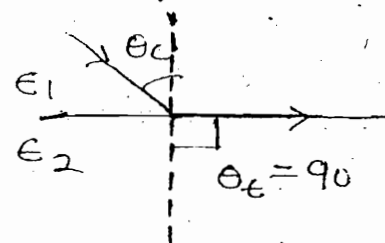
→ Refracting Index

Snell's 1st law $\rightarrow$

$$\theta_i = \theta_r$$

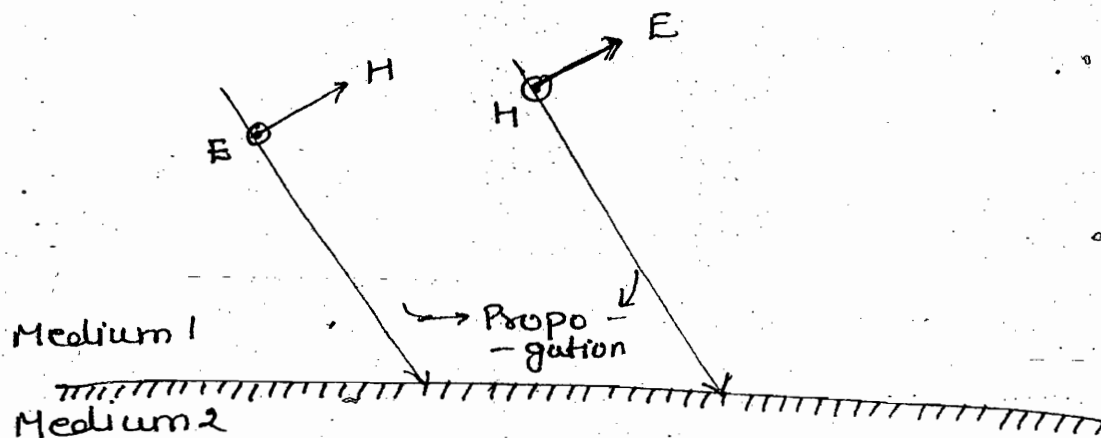
If $\theta_t = 90^\circ$ then $\theta_i = \theta_c = \text{critical angle}$

For all $\theta_i > \theta_c$ complete reflections and zero transmission takes place



$$\sin \theta_c = \sqrt{\frac{\epsilon_2}{\epsilon_1}}$$

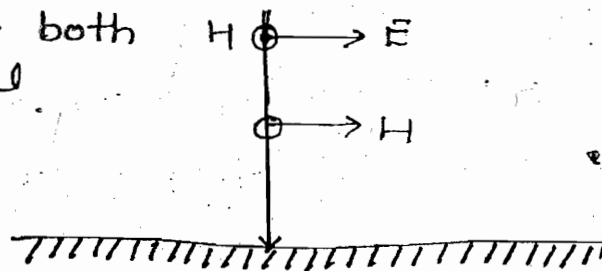
S & P Polarized Waves:-



S-Polarized Horizontal Polarization

P-Polarized vertical polarization.

In normal incidence both fields are Horizontal



Applying the Boundary Conditions,

$$E_{t1} = E_{t2}$$

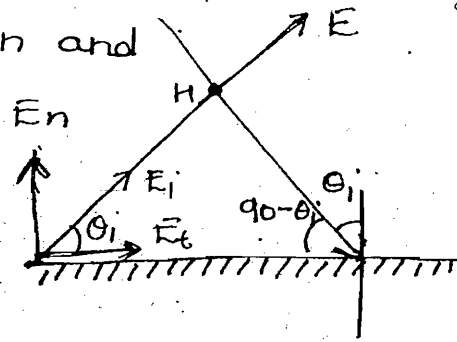
For s-polarized waves

$$E_i + E_r = E_t$$

$$1 + \Gamma_s = \tau_s \quad \text{--- (1s)}$$

Note:-

The angle b/w propagation and normals is the same angle b/w fields and surface



For p-polarised waves

$$E_i \cos \theta_i + E_r \cos \theta_r = E_t \cos \theta_t$$

$$E_i + E_r = E_t \frac{\cos \theta_t}{\cos \theta_i}$$

$$\boxed{1 + \Gamma = \tau \frac{\cos \theta_t}{\cos \theta_i}} \quad \text{--- } 1p$$

Similarly using $H_{t1} = H_{t2}$

$$\Gamma_s = \frac{n_2 \sec \theta_t - n_1 \sec \theta_i}{n_2 \sec \theta_t + n_1 \sec \theta_i}$$

$$\Gamma_p = \frac{n_2 \cos \theta_t - n_1 \cos \theta_i}{n_2 \cos \theta_t + n_1 \cos \theta_i}$$

Note:-

$\Gamma_s \neq \Gamma_p$ in oblique incidence

but when $\theta_i = \theta_r = \theta_t = 0$ then

$$\Gamma_s = \Gamma_p = \frac{n_2 - n_1}{n_2 + n_1}$$

$$\text{As } n_1 = \sqrt{\frac{\mu_0}{\epsilon_1}} \quad n_2 = \sqrt{\frac{\mu_0}{\epsilon_2}}$$

$$\Gamma_s = \frac{\sqrt{\epsilon_1} \cos \theta_i - \sqrt{\epsilon_2} \cos \theta_t}{\sqrt{\epsilon_1} \cos \theta_i + \sqrt{\epsilon_2} \cos \theta_t}$$

$$\frac{\sin \theta_i}{\sin \theta_t} = \sqrt{\frac{\epsilon_2}{\epsilon_1}}$$

$$\sin \theta_t = \sqrt{\frac{\epsilon_1}{\epsilon_2}} \sin \theta_i$$

$$\cos \theta_t = \sqrt{1 - \frac{\epsilon_1}{\epsilon_2} \sin^2 \theta_i}$$

$$\Gamma_s = \frac{\cos \theta_i - \sqrt{\frac{\epsilon_2}{\epsilon_1} - \sin^2 \theta_i}}{\cos \theta_i + \sqrt{\frac{\epsilon_2}{\epsilon_1} - \sin^2 \theta_i}}$$

$$\Gamma_p = \frac{\frac{\epsilon_2}{\epsilon_1} \cos \theta_i - \sqrt{\frac{\epsilon_2}{\epsilon_1} - \sin^2 \theta_i}}{\frac{\epsilon_2}{\epsilon_1} \cos \theta_i + \sqrt{\frac{\epsilon_2}{\epsilon_1} - \sin^2 \theta_i}}$$

If $\Gamma_p = 0$ i.e. zero reflections & complete transmission

$$\left(\frac{\epsilon_2}{\epsilon_1}\right)^2 (1 - \sin^2 \theta_i) = \frac{\epsilon_2}{\epsilon_1} - \sin^2 \theta_i$$

$$\sin^2 \theta_i \left(1 - \left(\frac{\epsilon_2}{\epsilon_1}\right)^2\right) = \frac{\epsilon_2}{\epsilon_1} - \left(\frac{\epsilon_2}{\epsilon_1}\right)^2$$

$$\sin^2 \theta_i = \frac{\frac{\epsilon_2}{\epsilon_1} \left(1 - \frac{\epsilon_2}{\epsilon_1}\right)}{\left(1 + \frac{\epsilon_2}{\epsilon_1}\right) \left(1 - \frac{\epsilon_2}{\epsilon_1}\right)}$$

$$\sin \theta_i = \sqrt{\frac{\epsilon_2}{\epsilon_1 + \epsilon_2}}$$

$$\tan \theta_i = \sqrt{\frac{\epsilon_2}{\epsilon_1}}$$

This $\theta_i = \theta_B$ is called as Brewster angle.
for p-polarised wave

$$\text{If } \Gamma_s = 0$$

$$\cos^2 \theta_i = \frac{\epsilon_2}{\epsilon_1} - \sin^2 \theta_i$$

$$\Rightarrow \epsilon_2 = \epsilon_1$$

Brewster angle and zero reflections cannot occur for s-polarized waves.

Summary:-

zero transmission, complete reflections

θ_c = critical angle

zero reflections, complete transmission

θ_B = Brewster angle

$$\sin \theta_c = \sqrt{\frac{\epsilon_2}{\epsilon_1}}$$

$$2. \tan \theta_B = \sqrt{\frac{\epsilon_2}{\epsilon_1}}$$

$\epsilon_1 > \epsilon_2$ is a required condition

3. No such medium restriction

No such polarization restriction

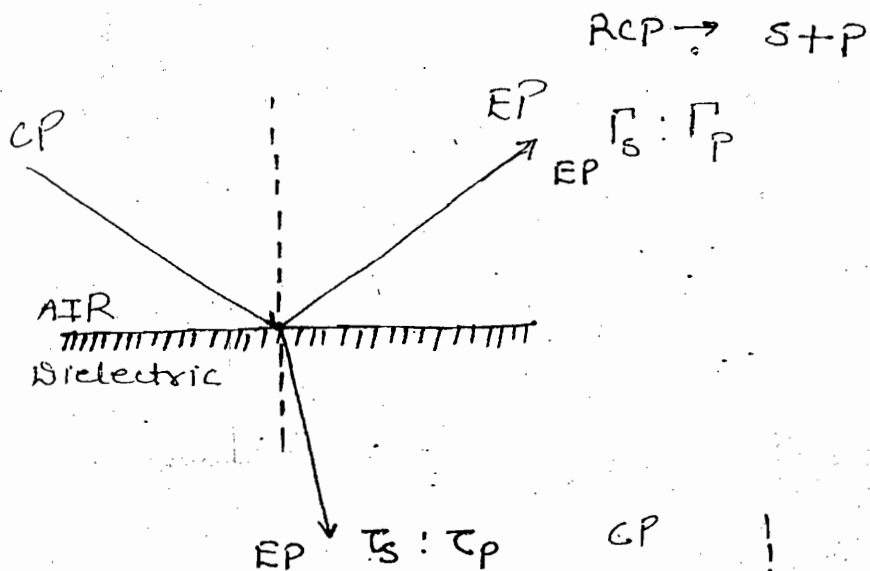
4. Exists only for p-polarized waves

For all $\theta_i > \theta_c$

→ the same effect

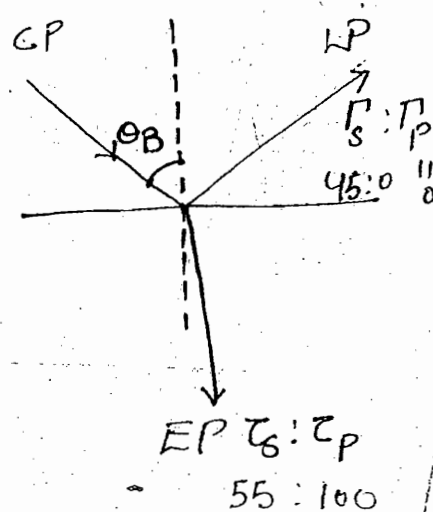
5. At $\theta_i = \theta_B$ only this occurs

19.



$$\tan \theta_B = \tan 60^\circ = \sqrt{\frac{\epsilon_R \epsilon_0}{\epsilon_0}}$$

$$\Rightarrow \epsilon_R = 3$$

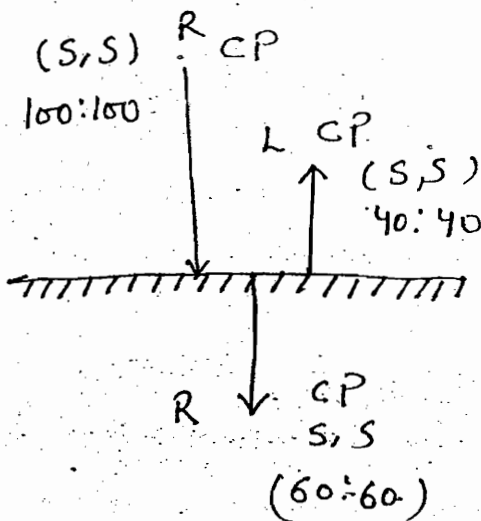


50

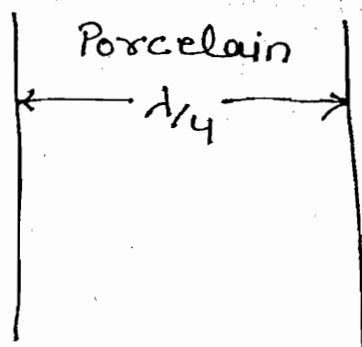
Linearly Polarised

51

a b



52



$$\text{Thickness} = \frac{\lambda}{4} = \left(\frac{3 \times 10^8}{\sqrt{\epsilon_R} \times 10^8 \times 4} \right)$$

53.

μ_R $\left\{ \begin{array}{l} J_b \rightarrow \text{Bond current density} \\ \rightarrow \text{Due to spinning/revolving electrons} \\ \text{constituting a magnetic dipole.} \end{array} \right.$

$\rightarrow [J_c \rightarrow \text{conduction current} - \text{moving electron}]$

$\in [J_d \rightarrow \text{displacement current} \rightarrow \text{changing Electric flux}]$

41.

$$\vec{E} = 3 \sin(\omega t - \beta z) \hat{a}_x + 6 \sin(\omega t - \beta z + 75^\circ) \hat{a}_y$$

$$E_0 = \sqrt{3^2 + 6^2} = 3\sqrt{5}$$

$$P_{avg} = \frac{1}{2} \frac{E_0^2}{\eta} \hat{a}_z$$

$$= \frac{1}{2} \frac{(3\sqrt{5})^2}{120\pi} \hat{a}_z \approx 58 \text{ mW}$$

OR

$$\vec{E} = 3 \sin(\omega t - \beta z) (\hat{a}_x + 2e^{j75^\circ} \hat{a}_y)$$

$$\vec{H} = \left(\frac{3}{120\pi} \right) \sin(\omega t - \beta z) (\hat{a}_y - 2e^{j75^\circ} \hat{a}_x)$$

$$P_{avg} = \frac{1}{2} \cdot 3 (\hat{a}_x + 2e^{j75^\circ} \hat{a}_y) \times \left(\frac{3}{120\pi} (\hat{a}_y - 2e^{-j75^\circ} \hat{a}_x) \right)$$

$$= \frac{1}{2} \frac{3 \cdot 3}{120\pi} (\hat{a}_z + 4\hat{a}_z)$$

$$= \frac{1}{2} \frac{(3\sqrt{5})^2}{120\pi} \hat{a}_z$$